Borehole 2 renewal: apply for 1,300 cubic metres a day, not 1,800

Francis Mason, 13 November 2026. Licence MA/28/15/0042. For the application due 18 December 2026.

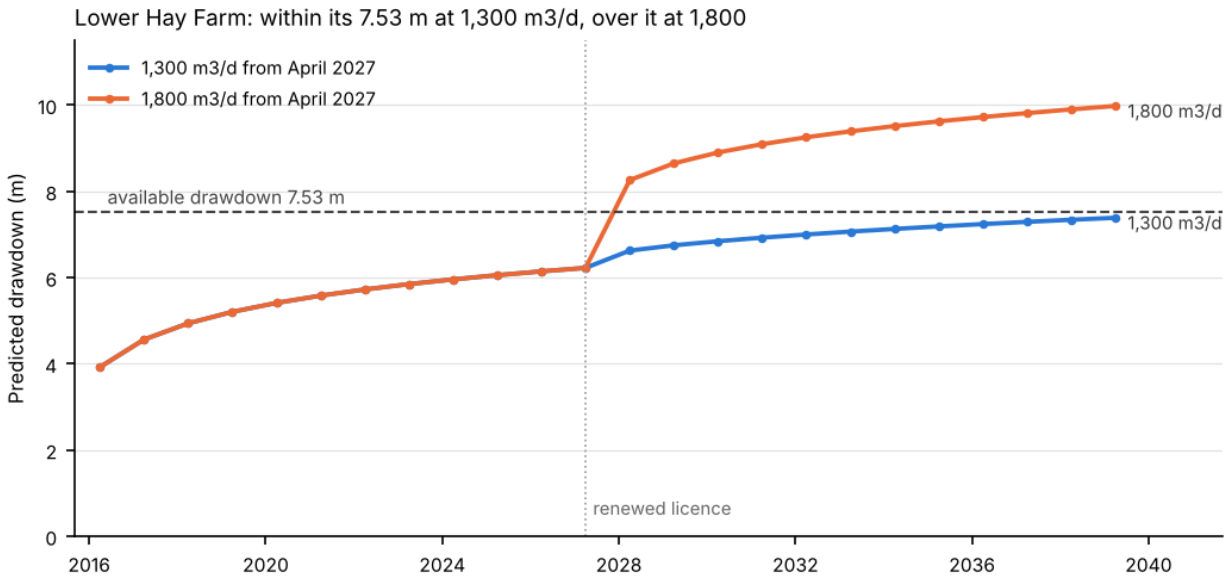
Apply for 1,300 cubic metres a day. The 1,800 a day line 5 needs fails the licensing team's derogation test at all three protected boreholes. Lower Hay Farm limits the quantity: 7.4 m predicted at 1,300 a day against its 7.53 m available drawdown.

Derogation test at 31 March 2039

Protected borehole	Available drawdown (m)	Predicted at 1,800 (m)	Predicted at 1,300 (m)	Passes at 1,800
Lower Hay Farm	7.53	10	7.4	no
Kilnbrook Nurseries	10.00	11	8.1	no
Hollin Bank Golf Club	9.90	10	7.4	no

Available drawdowns as tabled in the licensing team's letter of 22 September 2026. Drawdowns at 1,800 to the nearest metre, at 1,300 to one decimal.

Basis: Borehole 2 at 1,200 m3/d from 1 April 2015 and at the applied quantity from 1 April 2027, drawdowns from each borehole's April 2015 rest level (the letter's test). Aquifer: transmissivity 206 m2/d and storage coefficient 0.00036 from the June 2014 constant-rate test (all three observation boreholes), with the Coldharbour Fault as a no-flow boundary: it throws the Sherwood Sandstone against Mercia Mudstone, so the cone is held at the fault and each protected borehole also carries the drawdown of an image of Borehole 2 across it. The 72-hour test never reached the fault; the observation boreholes' record from 2015 to March 2026 (OB2 read from its raised casing top from March 2020), less the regional movement at Windle Cross and under the returned abstraction, matches this model to 0.03 m (rms) against 0.31 m for an unbounded aquifer, and Windle Cross, beyond the fault, shows no response.



Lower Hay Farm's predicted drawdown at each 31 March under the two quantities, against its available drawdown.

Need: line 3 water use, 2025-26

Line 5 copies line 3, so the application's need is sized on line 3's water use per litre packed, its cleans included (meter FM-L3, packed output from the MES at the bottle sizes filled that year: the 75cl still glass was still 750 ml until April 2026).

Month	Litres of water per litre packed
April 2025	2.38
May 2025	2.32
June 2025	2.38
July 2025	2.33
August 2025	2.40
September 2025	2.33
October 2025	2.39
November 2025	2.34
December 2025	2.35

Month	Litres of water per litre packed
January 2026	2.37
February 2026	2.36
March 2026	2.33

FM-L3's register from the site telemetry by local month, over the good bottles in the MES orders at the size filled.

Returns check

The bulk bay that opened for the Ferncastle Foods contract in April 2026 draws off the raw main upstream of FM-1, so the tanker loads are taken from Borehole 2 without passing the licence meter and FM-1 understates this year's abstraction. Each load is matched to its pump run in the wellhead log (the haulier dates its tickets by the night shift). Monthly volumes and days over 1,200 m3 from April to September 2026 are in bh2_returns_check.csv; the 2026-27 return will need the bulk loads added.